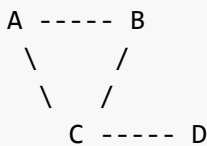


🌟 Graph Theory & Combinatorics Solutions: 2025

🌟 Q1 (i) Consider a graph with four vertices labeled A, B, C and D. The graph contains the following undirected edges: A connected to B, A connected to C, B connected to C, C connected to D. Find the corresponding graph structure, adjacency matrix, incidence matrix, and path matrix of length 2.

🌟 **Answer:**

♦ Step 1: Draw the Graph



The edges are: AB, AC, BC, and CD.

🌟 Step 2: Adjacency Matrix

♦ **Definition:** The adjacency matrix rows and columns represent vertices. An entry is **1** if an edge exists between those two vertices, and **0** otherwise.

🔥 **Vertex Order:** A, B, C, D

🔥 **Adjacency Matrix M :** $M = \begin{bmatrix} 0 & 1 & 1 & 0 \\ 1 & 0 & 1 & 0 \\ 1 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix}$

🌟 Explanation:

- $M_{12} = 1$ because A is directly connected to B.
- $M_{14} = 0$ because A is not connected to D.

🌟 Step 3: Incidence Matrix

♦ **Definition:** Rows represent vertices and columns represent edges. The entry is **1** if the vertex is incident to the edge, and **0** otherwise.

🔥 Edge Labels:

- $e_1 = AB$
- $e_2 = AC$
- $e_3 = BC$
- $e_4 = CD$

 **Incidence Matrix I :** $I = \begin{bmatrix} 1 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 \\ 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 1 \end{bmatrix}$

 **Step 4: Path Matrix of Length 2** The number of paths of length 2 is obtained directly from computing M^2 :

$$M^2 = \begin{bmatrix} 2 & 1 & 1 & 1 \\ 1 & 2 & 1 & 1 \\ 1 & 1 & 3 & 0 \\ 1 & 1 & 0 & 1 \end{bmatrix}$$

 **Interpretation:**

- Entry $(1,4) = 1$ means:  There is exactly one path of length 2 from A to D. That path is $A \rightarrow C \rightarrow D$.

 **Final Answer:**

Adjacency Matrix: $\begin{bmatrix} 0 & 1 & 1 & 0 \\ 1 & 0 & 1 & 0 \\ 1 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix}$

Incidence Matrix: $\begin{bmatrix} 1 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 \\ 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 1 \end{bmatrix}$

Path Matrix of Length 2: $\begin{bmatrix} 2 & 1 & 1 & 1 \\ 1 & 2 & 1 & 1 \\ 1 & 1 & 3 & 0 \\ 1 & 1 & 0 & 1 \end{bmatrix}$

 **Q1 (ii)** Prove that "If a graph has exactly two vertices of odd degrees, there must be a path joining these two vertices."

 **Answer:**

♦ **Important Theorem:** In every graph, the number of odd-degree vertices is always even. This fundamentally follows from the Handshaking Lemma.

♦ **Given:** The graph contains exactly two odd-degree vertices. Let them be u and v .

💡 **Proof:** Suppose no path exists between u and v . This means that u and v must belong to entirely different connected components of the graph.

 **Apply Odd Degree Property to Components:** In every individual connected component, the number of odd-degree vertices must be an even number. But under our assumption:

- One component contains only u (exactly **1** odd-degree vertex).
- Another component contains only v (exactly **1** odd-degree vertex).

Each of these components now has an odd number of odd-degree vertices, which is completely impossible!

✗ Odd-degree vertices must always occur in pairs within any component.

 **Therefore:** The assumption is false, and there must exist a valid path joining u and v .

 **Suitable Example:**

A ----- B ----- C

Degrees:

- Vertex A = 1
- Vertex B = 2
- Vertex C = 1

The odd-degree vertices are A and C. There clearly exists a path $A \rightarrow B \rightarrow C$ joining them. Hence, the theorem is verified! ✓

 **Final Conclusion:** If a graph has exactly two odd-degree vertices, then there must be a path connecting them.

 **Q1 (iii)** Find the maximum and minimum number of edges in a 3-component graph having 10 vertices.

 **Answer:**

◆ **Given:**

- Number of vertices: $n = 10$
- Number of components: $k = 3$

 **Minimum Number of Edges** For minimum edges, each component should be a simple tree structure. We know the formula for a graph with k components is: $|E|_{\min} = n - k$

Substitute the given values: $10 - 3 = 7$

✓ Minimum Edges =  **7**

 **Maximum Number of Edges** To maximize edges, we make one component as large a complete graph as possible, leaving the remaining components as isolated single vertices.

Largest component size: $10 - (3 - 1) = 8$ vertices.

The maximum number of edges in a complete graph of 8 vertices is: $\frac{8(7)}{2} = 28$

✓ Maximum Edges =  **28**

 **Final Answer:**

Quantity	Value
Minimum edges	7
Maximum edges	28

 **OR Question:** In a group of twelve persons, is it possible that each person has friendship with exactly five others in the group? Justify this.

✨ Answer:

◆ Represent as a Graph:

- Each person = vertex 🧑
- Friendship = edge 🤝

Given:

- Total persons (n) = **12**
- Each person has a degree of **5**.

🔥 **Total Degree Sum:** $12 \times 5 = 60$

◆ **Apply Handshaking Lemma:** We know that the sum of degrees must always be an even number because it equals $2|E|$. Here, **60** is an **EVEN** number ✅. Hence, such a graph is completely possible!

✅ **Final Answer:** ✨ Yes, it is perfectly possible because the total degree sum equals **60**, which is an even number, fully satisfying the Handshaking Lemma.

☀️ Q3 (i-a) Perform the union operation on G_1 and G_2 .

✨ Answer:

◆ Given Graphs:

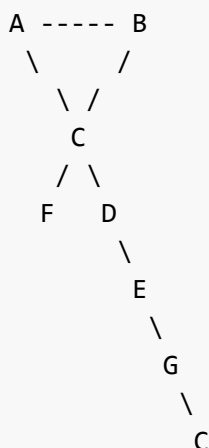
- **Graph G_1 :** Vertices $V_1 = \{A, B, C, F\}$, Edges $E_1 = \{AB, AC, BC, CF\}$
- **Graph G_2 :** Vertices $V_2 = \{C, D, E, G\}$, Edges $E_2 = \{CD, DE, EG, GC\}$

🔥 **Union of Graphs:** The union operation combines all unique vertices and all unique edges from both graphs together.

☀️ **Vertex Set:** $V(G_1 \cup G_2) = V_1 \cup V_2 = \{A, B, C, D, E, F, G\}$

☀️ **Edge Set:** $E(G_1 \cup G_2) = E_1 \cup E_2 = \{AB, AC, BC, CF, CD, DE, EG, GC\}$

🌈 Resulting Structure:



✓ **Final Answer:**

- **Vertices:** $\{A, B, C, D, E, F, G\}$
- **Edges:** $\{AB, AC, BC, CF, CD, DE, EG, GC\}$

☀️ Q3 (i-b) From the union graph, delete vertex D.

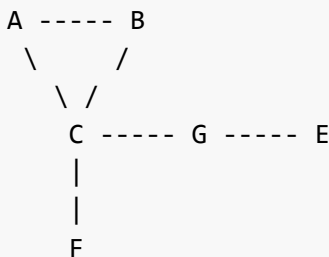
✨ **Answer:**

♦ **Deleting a Vertex:** Deleting vertex D means removing vertex D from the vertex set and removing all edges incident on it.

🔥 **Edges Incident on D:** The edges CD and DE are completely removed.

☀️ **Remaining Vertices:** $\{A, B, C, E, F, G\}$ ☀️ **Remaining Edges:** $\{AB, AC, BC, CF, EG, GC\}$

🌈 **Resulting Graph:**



✓ **Final Answer:** After deleting vertex D:

- **Vertices:** $\{A, B, C, E, F, G\}$
- **Edges:** $\{AB, AC, BC, CF, EG, GC\}$

☀️ Q3 (i-c) Identify at least one cut vertex and one cut edge in the union graph, and justify how their removal affects the graph.

✨ **Answer:**

♦ **Cut Vertex** A cut vertex is a vertex whose removal strictly increases the number of connected components.

🔥 **Identify Cut Vertex:** Consider vertex **C**. If vertex C is removed, the left part $\{A, B\}$ becomes completely disconnected from the right part $\{D, E, G, F\}$. Hence, **C** is a cut vertex. ✓

♦ **Effect of Removing C:** The graph breaks apart into separate, disconnected components.

♦ **Cut Edge** A cut edge (bridge) is an edge whose removal disconnects the graph.

🔥 **Identify Cut Edge:** Consider edge **CF**. If edge CF is removed, vertex F becomes completely isolated from the rest of the graph. Hence, **CF** is a cut edge. ✓

◆ **Effect of Removing CF:** The graph becomes disconnected because vertex F completely loses its path to all other vertices.

✓ **Final Answer:**

Type	Element
Cut Vertex	C
Cut Edge	CF

☀️ Q3 (i-d) Perform a fusion operation by merging vertices A and D, and update the graph accordingly.

✨ **Answer:**

◆ **Fusion Operation:** Fusion means combining two vertices into a single new vertex, where all edges originally incident to either vertex become attached to the new merged vertex.

🔥 **Merge Vertices A and D:** Let the new merged vertex be labeled **AD**.

☀️ **Old Edge List:** AB, AC, BC, CF, CD, DE, EG, GC ☀️ **Updated Edge List:** Replacing A and D with AD gives us the final simplified edge set:

$\{\text{AD-B, AD-C, BC, CF, AD-E, EG, GC}\}$

🌈 **Updated Graph Structure:**



✓ **Final Answer:**

- **New Vertex Set:** $\{\text{AD, B, C, E, F, G}\}$
- **New Edge Set:** $\{\text{AD-B, AD-C, BC, CF, AD-E, EG, GC}\}$

☀️ Q3 (ii) "The vertex connectivity of any graph G can never exceed the edge connectivity of G ." Justify this.

✨ **Answer:**

◆ **Definitions:**

- **Vertex Connectivity $K(G)$** = minimum number of vertices whose removal disconnects the graph.
- **Edge Connectivity $\lambda(G)$** = minimum number of edges whose removal disconnects the graph.

 **Important Theorem:** For every connected graph: $K(G) \leq \lambda(G)$

 **Example: Complete Graph K_4**



 **Vertex Connectivity:** We need to remove at least **3** vertices to disconnect the remaining graph. Hence, $K(G) = 3$.

 **Edge Connectivity:** We need to remove at least **3** edges to disconnect the graph. Hence, $\lambda(G) = 3$.

 **Verification:** $K(G) = \lambda(G) = 3$, which perfectly satisfies $K(G) \leq \lambda(G)$. 

 **Why This Happens:** Removing a vertex automatically removes all of its incident edges as well. Therefore, disconnecting a graph using vertex removal is never harder than doing so strictly through edge removal.

 **Final Conclusion:** The vertex connectivity of any graph can never exceed its edge connectivity.

 **Q4 (ii)** Consider a graph $G=(V,E)$ where $V=\{A,B,C,D,E,F\}$ and $E=\{(A,B),(A,C),(A,D),(B,C),(C,D),(D,E),(E,F),(C,F)\}$. Find the chromatic number of the graph. Mention all steps properly.

 **Answer:**

 **Chromatic Number:** The chromatic number $\chi(G)$ is the minimum number of colors required to color the graph such that no two adjacent vertices share the same color.

 **Step 1: Observe the Graph Structure** The graph contains multiple triangles, such as $A-B-C$ and $A-C-D$. A triangle (a complete graph K_3) always requires a minimum of **3** unique colors. Therefore, $\chi(G) \geq 3$.

 **Step 2: Try Coloring with 3 Colors** Let's use the colors:  Red,  Blue,  Green.

 **Color Assignment:**

- **A:**  Red
- **B:**  Blue
- **C:**  Green
- **D:**  Blue (adjacent to A and C)
- **E:**  Red (adjacent to D)
- **F:**  Blue (adjacent to E and C)

 **Verification of Edges:** All edges connect vertices of entirely different colors!

- AB ( , ) — Valid! 
- AC ( , ) — Valid! 
- AD ( , ) — Valid! 
- BC ( , ) — Valid! 
- CD ( , ) — Valid! 
- DE ( , ) — Valid! 
- EF ( , ) — Valid! 
- CF ( , ) — Valid! 

Since a valid 3-coloring is successfully achieved and 2-coloring is impossible due to the odd cycles (triangles), the chromatic number is exactly 3.

 **Final Answer:** $\chi(G) = 3$ The chromatic number of the graph is:  **3**

 **Q4 (iii-a)** Draw the graph G given $V = \{A, B, C, D, E, F, G, H\}$ and $E = \{(A, B), (B, C), (C, D), (E, F), (F, G), (G, H)\}$.

 **Answer:**

 **Graph Structure:**

```
A ----- B ----- C ----- D
E ----- F ----- G ----- H
```

The graph consists of exactly two disconnected components, where each component is a simple standalone path graph.

 **Final Answer:** The graph consists of two completely separate linear paths: $A-B-C-D$ and $E-F-G-H$.

 **Q4 (iii-b)** Determine the maximum independent set of G .

 **Answer:**

◆ **Independent Set:** An independent set is a set of vertices in a graph such that no two vertices in the set are adjacent.

 **Observe the First Component ($A-B-C-D$):** The maximum independent set can select alternating vertices: $\{A, C\}$ or $\{B, D\}$. Maximum size = 2.

 **Observe the Second Component ($E-F-G-H$):** Similarly, we can select alternating vertices here: $\{E, G\}$ or $\{F, H\}$. Maximum size = 2.

 **Combine Both Components:** Combining the optimal selections from both components gives us a maximum independent set. One such set is: $\{A, C, E, G\}$.

✓ **Final Answer:** A maximum independent set of the graph is: 🌟 $\{A, C, E, G\}$

🌟 Q4 (iii-c) What is the size of this maximum independent set? Justify your answer.

🌟 **Answer:**

The size of the maximum independent set $\{A, C, E, G\}$ is exactly **4**.

🔍 **Justification:** Each separate component is a path graph with 4 vertices. For any path graph of length 4, the maximum number of non-adjacent vertices you can select is $\lceil 4/2 \rceil = 2$. Since the graph consists of two completely disconnected components, the total maximum size is the sum of the capacities of both parts: $2 + 2 = 4$

No larger independent set can possibly exist without picking adjacent items.

✓ **Final Answer:** 🌟 Size of the maximum independent set = **4**

🌟 Q4 (iv) Determine whether the graph is planar or non-planar using Kuratowski's Theorem. Given Vertices: $\{A, B, C, D, E\}$, Edges: $\{AB, AC, AD, BC, BD, CD, DE, CE\}$.

🌟 **Answer:**

♦ **Kuratowski's Theorem:** A graph is non-planar if and only if it contains a subgraph that is a subdivision of K_5 (complete graph on 5 vertices) or $K_{3,3}$ (complete bipartite graph on 6 vertices).

🔍 **Observe the Given Graph:**

- Vertices $\{A, B, C, D\}$ are completely interconnected, forming a K_4 structure (6 edges).
- Vertex E is connected to the graph only via edges CE and DE .

🔴 **Check for K_5 :** A complete K_5 graph requires **5** vertices where every single pair is connected (10 total edges). In this graph, vertex E is completely missing connections to vertex A and vertex B . It lacks the edges or edge-subdivisions required to form a K_5 homeomorph.

🔴 **Check for $K_{3,3}$:** A $K_{3,3}$ configuration strictly requires a minimum of **6** vertices. Since our graph only contains **5** vertices in total, it cannot contain a subdivision of $K_{3,3}$.

🌟 **Conclusion:** Since the graph does not contain any subdivision of K_5 or $K_{3,3}$, it does not violate Kuratowski's criterion.

✓ **Final Answer:** 🌟 The graph is **PLANAR**.

🌟 Q4 (v) Find out how many vertices can be matched using maximum matching in the bipartite graph given below (Figure 2).

🌟 **Answer:**

♦ **Maximum Matching:** A maximum matching is a matching that contains the largest possible number of edges with no shared endpoints.

🔍 **Observe Figure 2:** We can perfectly match all pairs from the top and bottom partitions using the following edge set:

- (a, w)
- (b, v)
- (c, x)
- (d, y)
- (e, z)

🔥 **Number of Matching Edges:** There are **5** completely independent matching edges. Each matching edge securely pairs exactly **2** vertices.

Total matched vertices: $5 \times 2 = 10$

✅ **Final Answer:** ☀️ Number of matched vertices = **10** (Since all available vertices are successfully matched, this is also a perfect matching!)

☀️ Q5 (i) How many license plates can be made using either: Three uppercase English letters followed by three digits OR Four uppercase English letters followed by two digits?

🌟 **Answer:**

☀️ **Case 1: Three Letters Followed by Three Digits (LLLDDD)**

- There are **26** uppercase English letters available for each letter slot.
- There are **10** digits (**0–9**) available for each digit slot.

Using the Product Rule: $26^3 \times 10^3 = 17,576 \times 1,000 = 17,576,000$

☀️ **Case 2: Four Letters Followed by Two Digits (LLLLDD)** Using the Product Rule: $26^4 \times 10^2 = 456,976 \times 100 = 45,697,600$

🔥 **Total License Plates (Sum Rule):** Since the choices are mutually exclusive ("OR"), we add the two totals together: $17,576,000 + 45,697,600 = 63,273,600$

✅ **Final Answer:** ☀️ Total number of license plates = **63,273,600**

☀️ Q5 (ii) How many cards must be selected from a standard deck of 52 cards to guarantee that at least three cards of the same suit are chosen?

🌟 **Answer:**

♦ **Use Pigeonhole Principle** 🐦 A standard deck contains 4 distinct suits: Hearts ♥, Diamonds ♦, Clubs ♣, and Spades ♠.

🔥 **Worst Case Scenario:** To delay getting 3 cards of the same suit for as long as possible, you draw exactly **2** cards from each of the 4 suits. $\text{Maximum draws without 3 of a suit} = 2 \times 4 = 8 \text{ cards}$

At this point, you hold 2 cards of each suit. The very next card you pick (**the 9th card**) must belong to one of the 4 suits, immediately creating a set of 3 cards of that suit.

✅ **Final Answer:** $8 + 1 = 9$ ☀️ Minimum cards required = **9**

☀️ Q5 (iii) A club has 25 members. How many ways are there to choose four members of the club to serve on an executive committee?

✨ **Answer:**

♦ **Important Observation:** We are choosing 4 members to serve together on a committee. No distinct roles or titles (like President or Secretary) are assigned, so the order of selection does not matter. This requires combinations.

🔥 **Formula:** $\binom{n}{r} = \frac{n!}{r!(n-r)!}$

Here, $n = 25$ and $r = 4$: $\binom{25}{4} = \frac{25 \times 24 \times 23 \times 22}{4 \times 3 \times 2 \times 1} = 12650$

✅ **Final Answer:** ☀️ Number of executive committees = **12,650**

☀️ Q5 (iv) There are six different candidates for governor of a state. In how many different orders can the names of the candidates be printed on a ballot?

✨ **Answer:**

♦ **Important Concept:** We are arranging 6 unique candidate names in a specific sequence. Since the exact arrangement order matters, this is a permutation problem.

🔥 **Number of Arrangements:** The number of ways to arrange n distinct objects in a line is given by $n!$: $6! = 6 \times 5 \times 4 \times 3 \times 2 \times 1 = 720$

✅ **Final Answer:** ☀️ Different ballot orders = **720**

☀️ Q6 (i) Find the generating function of the sequence in closed form: $\langle 1, a, a^2, a^3, \dots \rangle$

✨ **Answer:**

♦ **Observe the Sequence:** The terms represent a standard geometric series with a common ratio of ax when mapped to power series coefficients.

🔥 **Generating Function Expression:** $G(x) = 1 + ax + a^2x^2 + a^3x^3 + \dots$

We can rewrite this by grouping the powers: $G(x) = 1 + (ax) + (ax)^2 + (ax)^3 + \dots$

This is a classic infinite geometric series with a first term of **1** and a common ratio of **ax** . Using the sum formula $\frac{1}{1-r}$: $G(x) = \frac{1}{1-ax}$

✅ **Final Answer:** $\frac{1}{1-ax}$

🌟 Q6 (ii) Find the solution of the recurrence relation: $a_n = a_{n-1} + 2a_{n-2}$ with $a_0 = 2, a_1 = 7$.

🌟 **Answer:**

💠 **Step 1: Form the Characteristic Equation** Rewrite the recurrence relation with all terms shifted to one side: $a_n - a_{n-1} - 2a_{n-2} = 0$

This gives us the characteristic quadratic equation: $r^2 - r - 2 = 0$

🔥 **Step 2: Solve the Quadratic Equation** Factorize the quadratic expression: $(r - 2)(r + 1) = 0$

The distinct roots are: $r = 2, \quad r = -1$

🔥 **Step 3: State the General Solution** Since the roots are completely distinct, the general solution takes the form: $a_n = A(2^n) + B(-1)^n$

🔥 **Step 4: Solve for Constants Using Initial Conditions**

- For $a_0 = 2$: $A + B = 2$
- For $a_1 = 7$: $2A - B = 7$

Add the two systems of equations together to eliminate B : $(A + B) + (2A - B) = 2 + 7$ $3A = 9 \implies A = 3$

Now, substitute $A = 3$ back to solve for B : $3 + B = 2 \implies B = -1$

🔥 **Step 5: Formulate the Final Explicit Equation** Substitute the solved constants $A = 3$ and $B = -1$ back into the general expression: $a_n = 3(2^n) - (-1)^n$

🌈 **Verification:**

- For $n=0$: $3(1) - 1 = 2$ (Correct! ✅)
- For $n=1$: $3(2) - (-1) = 6 + 1 = 7$ (Correct! ✅)

🎯 **Final Conclusion:** The explicit closed-form solution to the recurrence relation is: 🌟 $a_n = 3(2^n) - (-1)^n$